

Food, Health and Nutrition

Science

May, 2026

Short Title:	Food, Health and Nutrition	
Faculty	Science and Computing	
Department	Science	
Cluster	Food Science	
Author	JGERAGHTY	
Credits	10	Level: Postgraduate

Description of Module / Aims

This course is designed for postgraduate students who have knowledge and experience in the land sciences and food production, processing, and marketing. The module will investigate human nutrition requirements including the role of macronutrients and micronutrients as part of healthy balanced diets. There will be a review of the scientific evidence and trends in nutrient content of common foods. The role of key nutrition requirements for healthy lifestyles will be explained for nutritional related diseases. The objective of this course is for graduates to be conversant with human dietary requirements, current food eating trends, and the critical need for balanced healthy diets.

Programmes

SCIE-0083	MSc in Organic and Biological Agriculture (WD_SOBIA_R)	1 / 2 / M
SCIE-0083	Postgraduate Programmes in Science (WD_SGRAD_XX)	5 / 0 / E

Indicative Content

- Human food dietary requirements for balanced nutrition
- Dietary behaviour, factors affecting food choice, e.g. social and cultural factors, food labelling
- Trends and developments to optimise health, non-nutrients, e.g. phytochemicals, prebiotics, functional foods, biofortification, GE food ingredients, personalised nutrition, nutrigenomics
- Human illnesses and diseases associated with poor nutrition (diabetes, colon cancer, obesity etc.)
- Food safety from a microbiological and toxicological perspective with an emphasis on organic food
- National policy and regulation on food safety relating to organic food production and sale

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate a detailed knowledge and comprehension of balanced nutrition requirements in the human diet.
2. Assemble the relevant factors and give examples of influences affecting human dietary behaviour.
3. Describe and evaluate the role of various non-nutrient ingredients in healthy human diets.
4. Explain and illustrate the vital role of nutrition in the aetiology of dietary related human diseases.
5. Demonstrate a detailed knowledge, and categorize the application, of food safety measures in relation to production and processing with an emphasis on organic food.
6. Appraise and critique contemporary published evidence and findings for developing human health and nutrition-related trends.
7. Select, interpret, and summarize relevant research and health study information on food, health and nutrition from a variety of literature and scientific sources.

Learning and Teaching Methods

- Online lectures

- Guest lectures and webinars
- Assignments and case studies for class discussion and self-directed learning

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5,6,7
<i>Assignment</i>	50%	1,2,3,4,5,6,7
<i>Case Studies</i>	35%	4,5,7
<i>Presentation</i>	15%	3,5,7

Assessment Criteria

- <40%:** Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.
- 40%--59%:** The learner will be able to demonstrate a good understanding of human nutrition requirements for human health and welfare. The student will be able to write and present a topical case study.
- 60%--69%:** In addition to the above, the student will demonstrate more detailed knowledge of all the module topics covered, including human health developments and trends. The student will also be able to write and present a good quality case study.
- 70%--100%:** The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the vital role of nutrition in human health and diseases associated with poor nutrition and write and present a related high quality case study.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial		12
Online Delivery		24
Online Research		12
Independent Learning		222

Essential Material

- Smith, J., S. Gropper and T. Carr. *Advanced Nutrition and Human Metabolism*. 8th ed.. Belmont, CA, USA: Wadsworth Publishing Co Inc, 2021.

Supplementary Material

- "European Food Safety Authority." <https://www.efsa.europa.eu/en>
- "Food Safety Authority of Ireland." <https://www.fsai.ie/>